TOP 50 REPEATED CASE STUDIES: MATHEMATICS

Strategic High-Repeat Prediction Series for 2026 Boards

Created for Thunderstudy By Aashi Mathematics

A. APPLICATION OF DERIVATIVES (OPTIMIZATION)

1. Rectangular Field:

Maximize area for a given fixed perimeter. (Find dimensions x, y)

2. Box from Sheet:

Cutting squares x from corners of a 24x24 sheet to maximize volume.

3. Cylindrical Container:

Given volume V, find r and h for minimum surface area.

4. Open Water Tank:

Minimizing material cost for a tank with square base.

5. Profit Maximisation:

$P(x) = R(x) - C(x)$, find x for max profit.

6. Area Under Curve:

Maximize rectangular area inscribed in an ellipse/parabola.

7. Garden Fencing:

Three-sided fencing along a wall to maximize garden area.

8. Sector to Cone:

Maximize cone volume formed by folding a circular sector.

9. Wire Bending:

28m wire split into circle and square; find lengths for min total area.

10. Surface Area Container:

Fixed surface area, maximize the holding capacity.

B. VECTORS MOVEMENT CASE STUDIES

11. Friends Outing:

Three friends starting at O move to A, B, C; find resultant displacement.

12. Traveler Meeting:

Intersection of two vector paths $r = a + \lambda b$.

13. Drone Flight:

Position vector path of a drone monitoring a site.

14. Boat Navigation:

Velocity and displacement vectors in river current.

15. Position Vectors:

Relation between vectors OA, OB, and OC for a triangle.

16. Path Angles:

Finding the cosine of the angle between two moving cars.

17. Unit Vector Perpendicular:

Finding direction of a third object perpendicular to two paths.

18. Resultant Displacement:

Calculating total distance from origin after multiple moves.

19. Distance Between Objects:

Magnitude of (Vector A - Vector B) at time t.

20. Collinearity:

Verifying if three landmarks are in a straight line using vectors.

C. DIFFERENTIAL EQUATION MODELS

21. Population growth:

$dP/dt = kP$, solve for P at time t.

22. Radioactive decay:

Rate proportional to substance remaining.

23. Newton's Law of Cooling:

$dT/dt = -k(T - T_s)$.

24. Bacterial culture:

Growth proportional to initial size.

25. Volume Growth:

Rate of change of volume of a spherical balloon.

26. Drug Metabolism:

Rate of decay of medicine in bloodstream.

27. Water Tank Model:

Inflow vs Outflow rate differential equation.

28. Compound Interest:

Rate of change of principal proportional to amount.

D. PROBABILITY REAL-LIFE MODELS

29. Mixed Selection:

Selecting cards and tossing coins conditionally.

30. Student Selection:

Selecting monitors with specified gender counts.

31. Factory Defects:

Probability of defective item from Machine A/B (Bayes').

32. Medical Tests:

Accuracy of disease detection tests.

33. Public Survey:

Opinions on social issues (Probability distribution).

34. E-Commerce Shopping:

Probability of cart checkout vs abandonment.

35. Match Winning:

Probabilities of Tie, Win, or Loss in sports.

36. Replacement Analysis:

Drawing balls from bags with/without replacement.

E. LINEAR PROGRAMMING (LPP) CASES

37. Factory Production:

Product X and Y with labor constraints.

38. Diet Problem:

Nutrition requirements for Vitamin A and B within budget.

39. Transport Cost:

Shipping from two godowns to three outlets.

40. Furniture Maker:

Optimal chairs/tables for max profit.

41. Resource Allocation:

Maximizing production given limited materials.

42. Farmer Crop Planning:

Selecting Wheat vs Maize land allocation.

F. INTEGRALS & GEOMETRY

43. Boundary Area:

Land area between a river curve and straight road.

44. Archway Area:

Area of a parabolic tunnel cross-section.

45. Water Discharge:

Finding cross-sectional area of water flow.

46. Road Design:

Intersection area of two roads (overlapping regions).

47. Shaded Logo:

Calculating area of a circular/elliptical logo segment.

G. RELATIONS & FUNCTIONS

48. Library Mapping:

Relation R in a set of books with same author.

49. Seating Arrangement:

Bijective check for students allocated to seats.

50. Human Relations:

Checking equivalence (Reflexive/Symmetric/Transitive).

⭐ WHAT YOUR PAPERS PROVE

Set 65/1/1: Focus on Solar Optimisation & Seating Relations.
Set 65/5/1: Geometry mapping & Vector movement scenarios.
Set 65/7/1: Growth/Decay models & Bayes' survey situations.

2026 Prediction: Prepare Optimization (Box/Cylinder), Vector Movement stories, and Bayes' survey situations for 90% coverage.